AN INVERSE SYSTEM OF QUASI-COMPACT TOPOLOGICAL SPACES WHOSE LIMIT IS NOT QUASI-COMPACT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof adds no mathematics of its own; the singleton covering having no finite subcover is the context's closing observation (333-335).

Editorial note. The label lemma-lim-not-quasi-compact names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 296-335: for each n the topology T_n on N with basis {1}, ..., {n}, I_n where I_n = naturals > n; each X_n = (N, T_n) is quasi-compact and the identity maps f_{n,m}: X_n -> X_m are continuous with the compatibilities f_{p,n} compose f_{n,m} = f_{p,m}, giving a directed inverse system of quasi-compact spaces
- Context continues: X = (N, discrete) with f_n: X -> X_n continuous; the universal property check (any compatible system of maps g_n: (Y, S) -> (N, T_n) is a common set map g: Y -> N with all g^{-1}({n}) open in Y, hence continuous for S and T) shows X is the inverse limit; finally the infinite open covering of X by singleton sets has no inverse limit, so X is not quasi-compact

Let $\mathbf{N}$ denote the set of natural numbers. For every integer n, let I_n denote the set of all natural numbers $> n$. Define T_n to be the unique topology on $\mathbf{N}$ with basis $\{1\}, \ldots, \{n\}, I_n$. Denote by X_n the topological space $(\mathbf{N}, T_n)$. For each $m < n$, the identity map,

$$f_{n,m} : X_n \longrightarrow X_m$$

is continuous. Obviously for $m < n < p$, the composition $f_{p,n} \circ f_{n,m}$ equals $f_{p,m}$. So $((X_n), (f_{n,m}))$ is a directed inverse system of quasi-compact topological spaces.

Let T be the discrete topology on $\mathbf{N}$, and let X be $(\mathbf{N}, T)$. Then for every integer n, the identity map,

$$f_n : X \longrightarrow X_n$$

is continuous. We claim that this is the inverse limit of the directed system above. Let (Y, S) be any topological space. For every integer n, let

$$g_n : (Y, S) \longrightarrow (\mathbf{N}, T_n)$$

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be a continuous map. Assume that for every $m < n$ we have $f_{n,m} \circ g_n = g_m$, i.e., the system (g_n) is compatible with the directed system above. In particular, all of the set maps g_n are equal to a common set map

$$g : Y \longrightarrow \mathbf{N}.$$

Moreover, for every integer n, since $\{n\}$ is open in X_n, also $g^{-1}(\{n\}) = g_n^{-1}(\{n\})$ is open in Y. Therefore the set map g is continuous for the topology S on Y and the topology T on $\mathbf{N}$. Thus $(X, (f_n))$ is the inverse limit of the directed system above.

However, clearly X is not quasi-compact, since the infinite open covering by singleton sets has no inverse limit.

Lemma 0.1. *There exists an inverse system of quasi-compact topological spaces over* $\mathbf{N}$ *whose limit is not quasi-compact.*

Proof. See discussion above. □